COURSE OUTLINE

2025/2026

COURSE NAME: Database Fundamentals
COURSE CODE: SDEV1201

COURSE DESCRIPTION

Databases and data are at the core of all businesses. This course will comprehensively cover how to design, build, and manage relational database systems. Students will learn how to create, update, and delete data. They will also learn how to select data using methods such as sorting, grouping, and filtering. Advanced topics will cover procedures, constraints, triggers, and transactions in databases.

Course Credits: 3.00
Equivalent Courses: DMIT1508

LEARNING OUTCOMES

OUTCOME	Upon successful completion of this course, you will be able to
1	Analyze an organization's data/business rules and design third normal form (3NF) entities utilizing the normalization process and industry best practices to reduce data anomalies and create an efficient, consistent, and effective entity-relationship model. The following concepts, skills, and issues are used to support this Outcome: • Understand the impact of data anomalies and redundancy in a relational database • Use the normalization process to design a relational database in third normal form • Create and interpret Entity Relationship Diagrams including cardinality and optionality symbols • Discuss various relational database management systems and their use cases
2	Write Structured Query Language (SQL) commands to create and alter tables, and add to, modify, and remove data in a relational database. The following concepts, skills, and issues are used to support this Outcome: • Learn the DDL (Data Definition Language) syntax for creating and altering tables in a relational database, including primary key, foreign key, check, default, and unique constraints • Use the DML (Data Manipulation Language) syntax to add, modify, and remove data • Incorporate best practices and use style guides when developing SQL • Discuss different database technologies

3	Create accurate and maintainable SQL queries to retrieve data from a relational database to satisfy users' information needs utilizing one or more tables and various functions. The following concepts, skills, and issues are used to support this Outcome: • Use SQL SELECT statements to query a database including sorting, filtering, and aggregating • Utilize debugging and testing strategies to check accuracy of queries • Describe the different types of JOINs • Use functions in SQL
4	Construct stored procedures including transactions and error handling within a relational database to improve security, performance, and code reusability. The following concepts, skills, and issues are used to support this Outcome: • Use decision structures • Implement error handling in stored procedures • Discuss debugging and testing strategies to verify logic
5	Discuss database triggers and how they automate actions and enforce complex business rules within a relational database. The following concepts, skills, and issues are used to support this Outcome: • Discuss use cases for trigger • Discuss how triggers enforce business rules
6	Discuss, analyze, and utilize different database systems and non-relational implementations. The following concepts, skills, and issues are used to support this Outcome: • Define structured vs non-structured data • Discuss non-relational implementations • Define big data and discuss how data sets are used as the basis for modelling & analytics

STUDENT EVALUATION

OUTCOME	ACTIVITY DESCRIPTION	MARK DISTRIBUTION
1	Take-home Assignment: Normalization	10%
1	In-Class Quiz: Entity-Relationship Diagram symbols & terminology	10%
2	Take-home Coding Assignment: CREATE TABLE & DML statements	5%
2	In-Class Quiz: CREATE TABLE & DML statements	10%
1, 2 and 6	Case Study: Video or Written Presentation on a database technology	15%
3	Take-home Coding Assignment: Basic SELECT statements	10%
3	Take-home Coding Assignment: Advanced SELECT statements	15%
3	In-Class Quiz: JOINs & SELECT statements	10%
2, 3, 4, 5 and 6	In-Class Quiz: Stored Procedures & Triggers	15%
	TOTAL	**100%**

COURSE COMPLETION REQUIREMENTS

To successfully complete this course, students must achieve an overall average of at least 50%.

STUDENT EQUIPMENT AND SUPPLIES

This program has implemented bring your own device (BYOD). Please ensure you have a laptop with the minimum specifications for your program. See the NAIT website for laptop requirements. All other equipment and supplies are listed in the course syllabus.

DELIVERY METHOD

This course will be taught using a variety of delivery methods which may include face-to-face, online, or blended teaching platforms. Activities such as collaborative exercises/assignments, seminars, labs, discussions, audio/visual presentations, and case studies may be used to support learning.

ADDITIONAL INFORMATION

All course assessments are expected to be completed before the specified deadline date and time. Extensions will only be considered for those students that have an accommodation letter from NAIT's Learning Services or are impacted by extenuating circumstances. Please be aware that students with accommodations will receive a maximum of 2 additional days on assessments. Please refer to your Brightspace course for further details regarding the program's assessment deadline policy.

STUDENT RESPONSIBILITY

It is expected that students will be responsible citizens of the Institute by following the Student Rights and Responsibilities Policy(SR 1.0). As such, each student will assist in the preservation of Institute property, and assume responsibility for their education by staying informed of and abiding by academic requirements and policies; demonstrating respect toward others; and meet expectations concerning attendance, assignments, deadlines, and appointments.

EQUITY AND INCLUSION STATEMENT

NAIT is committed to advancing equity and to actively and intentionally creating learning environments that promote a sense of belonging and dignity that ensure all people are safe, respected and valued. Acknowledging that every member of the NAIT community has a role in and responsibility to this work, NAIT provides the resources and support necessary for programs, departments and individuals to champion equity, diversity and inclusion and address barriers in meaningful ways.

TERRITORIAL ACKNOWLEDGEMENT

At NAIT, we honour and acknowledge that the land on which we learn, work and live is Treaty Six territory. We seek to learn from history and the lessons that have come before us, and to draw on the wisdom of the First Peoples in Canada. Only through learning can we move forward in truth and reconciliation, and to a better future together.

Leadership Review Date: October 29, 2024

Changes to This Course Outline: Every effort has been made to ensure that information in this course outline is accurate at the time of publication. The Institute reserves the right to change courses if it becomes necessary so that course content remains relevant. In such cases, the instructor will give the students clear and timely notice of the changes.

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